

Math 290

Practice Quiz 2 — Version G Solutions

1(a) Answer: C. The theorem is a universal statement, so checking a single pair (x, y) proves nothing. A correct proof by contradiction: let $x \in \mathbb{Q}$, $y \in \mathbb{R} - \mathbb{Q}$ and suppose $x + y = r \in \mathbb{Q}$. Then $y = r - x$ is a difference of rationals, hence rational (if $r = \frac{a}{b}$, $x = \frac{c}{d}$, then $r - x = \frac{ad-bc}{bd}$ with $bd \neq 0$), contradicting $y \notin \mathbb{Q}$. (Based on Exercise 9.6.)

1(b) Answer: True. At $x = 0$: $\frac{1}{2}x^2 = 0$ and $0 > 0$ is false, so $x = 0$ satisfies the negation $x \leq \frac{1}{2}x^2$. (Based on Section 11.D.)

1(c) Answer: False. The converse is “If $a \nmid b$ and $a \nmid c$, then $a \nmid bc$.” Counterexample: $a = 4$, $b = c = 2$. Then $4 \nmid 2$ twice, but $4 \mid 4 = bc$. (Based on Exercise 8.2.)

2. (Based on Exercises 7.5 and 7.6.)

Solution. Let $a, b, c, d \in \mathbb{Z}$. We work directly. Assume $a \mid c$ and $b \mid d$. Then $c = ak$ and $d = b\ell$ for some $k, \ell \in \mathbb{Z}$ (use two different letters!). So

$$cd = (ak)(b\ell) = (ab)(k\ell).$$

Since $k\ell \in \mathbb{Z}$, we have $ab \mid cd$.

Contrapositive: “If $ab \nmid cd$, then $a \nmid c$ or $b \nmid d$.” □

3. (Based on Exercise 11.1(c) and Proposition 11.7.)

Solution. Let $q = \sqrt{2}^{\sqrt{2}}$. We consider two cases.

Case 1: q is irrational. Fix $a = b = \sqrt{2}$, both irrational. Then $a^b = q$ is irrational.

Case 2: q is rational. Note $q > 0$, so $q \neq 0$ and $1/q$ is a nonzero rational number. Fix $a = \sqrt{2}$ and $b = 1 - \sqrt{2}$. Here b is irrational: $-\sqrt{2} = (-1)\sqrt{2}$ is a nonzero rational times an irrational, and then $1 + (-\sqrt{2})$ is a rational plus an irrational. By the hint,

$$a^b = \sqrt{2}^{1-\sqrt{2}} = \frac{\sqrt{2}}{\sqrt{2}^{\sqrt{2}}} = \frac{1}{q} \cdot \sqrt{2},$$

a nonzero rational times an irrational, so a^b is irrational.

In either case we have found irrational a, b with a^b irrational. (We don’t need to know which case actually holds — this is a nonconstructive proof.) □

4. (Based on Exercise 10.9 and Definition 2.9 (recall $\mathbb{N} = \{1, 2, 3, \dots\}$).)

Solution. (a) ($\subseteq$) Let $x \in \bigcup_{n \in \mathbb{N}} S_n$. Then $x \in S_n$ for some $n \in \mathbb{N}$, so $x \in \mathbb{Z}$ by the definition of S_n .

($\supseteq$) Let $x \in \mathbb{Z}$. Case 1: $x \leq 1$. Then $x \in S_1$. Case 2: $x > 1$. Then $x \in \mathbb{N}$ and $x \leq x$, so $x \in S_x$. In both cases $x \in S_n$ for some $n \in \mathbb{N}$, so $x \in \bigcup_{n \in \mathbb{N}} S_n$.

Hence $\bigcup_{n \in \mathbb{N}} S_n = \mathbb{Z}$.

(b) We claim $\bigcap_{n \in \mathbb{N}} S_n = S_1 = \{m \in \mathbb{Z} : m \leq 1\}$. ($\subseteq$) If x is in every S_n , then in particular $x \in S_1$. ($\supseteq$) If $x \in \mathbb{Z}$ and $x \leq 1$, then for every $n \in \mathbb{N}$ we have $x \leq 1 \leq n$, so $x \in S_n$. Thus $x \in \bigcap_{n \in \mathbb{N}} S_n$. □